

Style Transfer for Biometric Dataset Augmentation

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Task and Problem

- Goal** Face recognition in archival newsprint scans.
- Challenge** Printing artifacts (raster/noise) cause model failure.
- Solution** Synthetic data augmentation via Style Transfer (CUT).

Datasets and Pre-processing

Dataset	Domain	Images	IDs	Characteristics
WebFace4M	Source (Clean)	4,200,000	205,000	High-resolution modern portraits. Used for source distribution.
People Gator	Target (News)	16,000	9,180	Real scans from historical newspapers. High noise and printing raster.
WikiFace_112	Auxiliary	3,223	1,538	Historical studio portraits. Used for validation and style diversity.

Table 1. Experimental datasets overview

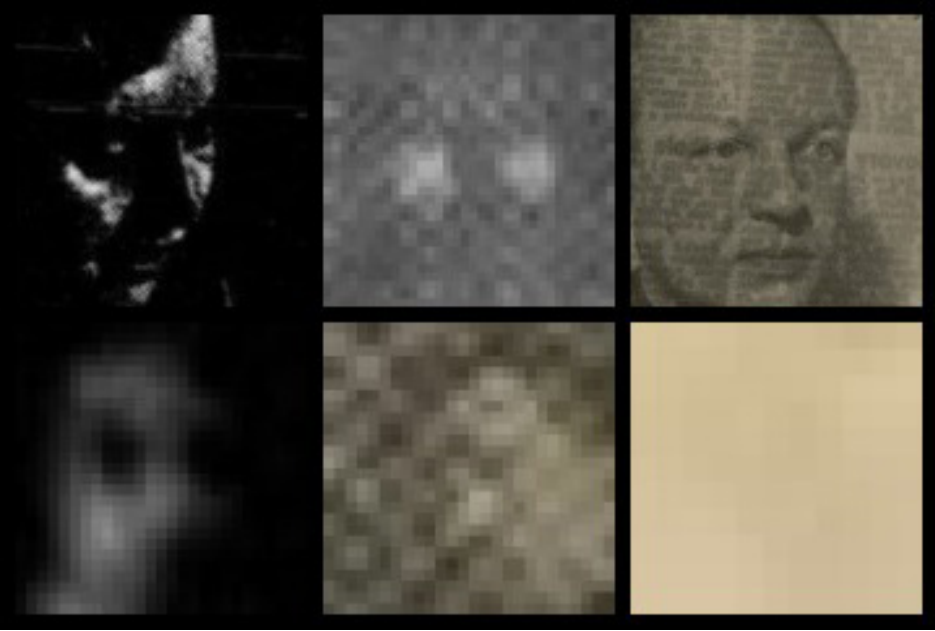


Figure 1. Dropped low-information samples

1. **Alignment:** 5-point ArcFace template mapping.
2. **Filtering:** Quality scoring + Haar face detection.
3. **Refinement:** Removing low-detail samples (down/up-scale test).
4. **Clustering:** Separating Sepia and Grayscale styles to prevent artifacts.

Style Transfer Methodology: CUT Model

Unpaired Learning; PatchNCE loss ensures facial geometry remains identical after transformation



Figure 2. Qualitative comparison across models. Split styles show the fewest artifacts and best noise stability

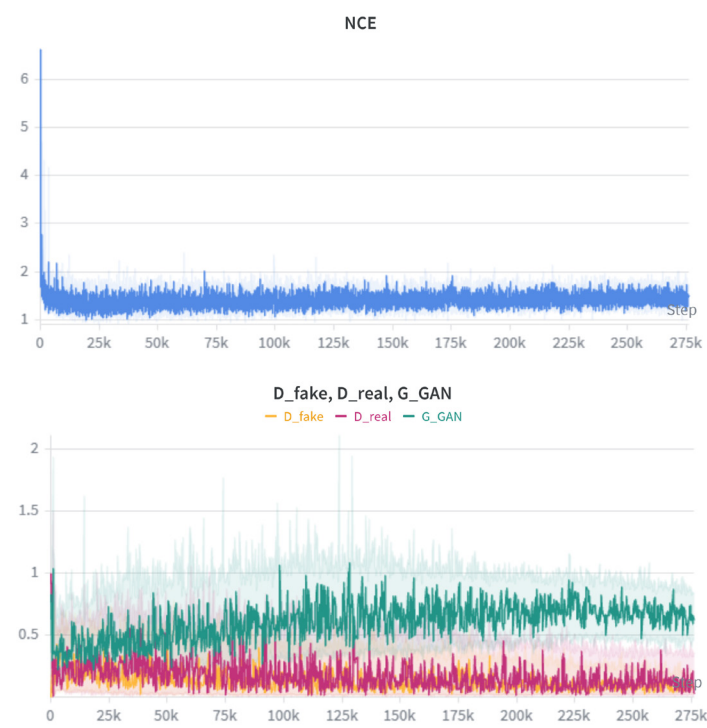


Figure 3. PatchNCE loss and Generator vs Discriminator min-max game

Results and Analysis

Top Performance: 0.788 P@1 (+7.0 pp vs. baseline); **Style Diversity:** Mixed styles > single domain
Color Matters: Sepia (pg-colored) > Grayscale; **Synthetic Utility:** Replaces missing archival data

Experiment	Strategy	R@1	R@5	TAR	P@1	NDCC@10	MRR@10
E0-dense-baseline [§]	clean, 2.5k IDs	0.234	0.434	0.199	0.718	0.591	0.758
E0-doubled[§]	clean, 5k IDs	0.337	0.549	0.137	0.788	0.650	0.821
E1-combined [†]	clean+CUT all	0.335	0.526	0.223	0.771	0.638	0.798
E1-people [†]	clean+CUT ppl	0.339	0.540	0.173	0.763	0.643	0.797
E1-pg-colored [†]	clean+CUT pgC	0.302	0.482	0.169	0.784	0.661	0.818
E1-pg-grayscale [†]	clean+CUT pgG	0.319	0.502	0.133	0.763	0.628	0.797
E1-wiki [†]	clean+CUT wiki	0.316	0.503	0.142	0.743	0.609	0.781
E4a-combined[‡]	CUT all (only)	0.327	0.523	0.170	0.788	0.650	0.821
E4b-people [‡]	CUT ppl	0.305	0.493	0.190	0.743	0.602	0.780
E4c-pgcolored [‡]	CUT pgC	0.330	0.506	0.165	0.763	0.640	0.803
E4d-pggrayscale [‡]	CUT pgG	0.271	0.456	0.153	0.759	0.619	0.795
E4e-wiki [‡]	CUT wiki	0.241	0.381	0.150	0.698	0.568	0.739

Table 2. Face recognition and retrieval performance metrics

- [§] **Baseline**
Clean data without augmentation.
- [†] **Augmented**
Clean data combined with CUT fake_B synthesis.
- [‡] **Synthetic**
Training using only CUT synthesized data.

Bold values indicate the best performance in the category.

[1] DENG, J., GUO, J., XUE, N. a ZAFEIRIOU, S. Arcface: Additive angular margin loss for deep face recognition.
[2] ISOLA, P., ZHU, J.-Y., ZHOU, T. a EFROS, A. A. Image-to-image translation with conditional adversarial networks.
[3] LIU, Z., MAO, H., WU, C., FEICHTENHOFER, C., DARRELL, T. et al. A ConvNet for the 2020s.
[4] PARK, T., EFROS, A. A., ZHANG, R. a ZHU, J.-Y. Contrastive learning for unpaired image-to-image translation.
[5] WANG, H., WANG, Y., ZHOU, Z., JI, X., LI, Z. et al. CosFace: Large Margin Cosine Loss for Deep Face Recognition.
[6] WANG, X., LI, Y., ZHANG, H. a SHAN, Y. Towards real-world blind face restoration with generative facial prior.
[7] ZHU, J.-Y., PARK, T., ISOLA, P. a EFROS, A. A. Unpaired image-to-image translation using cycleconsistent adversarial networks.
[8] ZHU, Z., HUANG, G., DENG, J., YE, Y., HUANG, J. et al. Webface260m: A benchmark unveiling the power of million-scale deep face recognition.